

SE4-DMAD — Exam Rehearsal Overview

Discrete Mathematics, Algorithms & Data Structures

One-week revision plan · ranked by exam value

1 How the exam actually works

The exam is a **digital multiple-choice test (MCQ)** taken on your own computer in SDU's "DE – Digital Eksamen" system. Key facts that shape the strategy below:

Format	Multiple choice. Questions shown one at a time; you can jump back and forth. Each question is worth a stated number of points (2–6 pts each).
Multi-answer	Many questions say " Et eller flere korrekte svar " (one <i>or more</i> correct answers) — you must tick every correct option and none of the wrong ones. These are where points are won and lost.
Aids	Limited aids allowed: textbooks, notes, lecture slides, formula sheets. No internet, no chatbots. So a good cheat-sheet of mechanical procedures is gold.
Reference	Pseudocode follows Cormen et al. (CLRS), 4th edition — algorithm names on the exam (PARTITION, BUILD-MIN-HEAP, RADIX-SORT, etc.) are CLRS names.

The single most important fact for prioritising: In the most recent combined exam (June 2025, 37 questions, ≈125 points total), **Algorithms & Data Structures was ≈80% of the points** and **Discrete Mathematics only ≈20%** (the last 5 questions, 25 pts). Spend your week accordingly — but note that the discrete-maths block is small, self-contained, and almost free points, so it must not be skipped.

2 The two halves of the course

2.1 Part A — Discrete Mathematics (≈20% of exam)

Compact: only **three chapters** (per the course notes, Noter 1–9). Almost every question is mechanical once you know the definitions.

Topic	CLRS/Rosen ch.	What you must be able to do
Logic	Ch. 1	Logical operators ($\neg \wedge \vee \Rightarrow \Leftrightarrow \oplus$) and precedence; truth tables ($\#rows = 2^{vars}$); equivalences (e.g. $p \Rightarrow q \equiv \neg p \vee q$, De Morgan); tautology / contradiction / contingency.
Quantifiers	Ch. 1.4–1.5	$\forall$ / $\exists$, nested quantifiers, <i>order matters</i> ($\forall x \exists y$ vs $\exists y \forall x$), negation of quantified statements.
Proof methods	Ch. 1 + 5	Direct proof, contraposition, proof by contradiction, induction (basis + induction step). Recognising a <i>valid</i> vs <i>broken</i> proof.
Relations	Ch. 9	Properties: reflexive, symmetric, antisymmetric, transitive; equivalence relation (refl+symm+trans); partial order (refl+antisymm+trans); reflexive/symmetric/transitive closures .

2.2 Part B — Algorithms & Data Structures (~80% of exam)

This is where the bulk of the points are. Grouped by theme below, with the slide deck that covers each.

Theme	Slides	Core skills tested
Asymptotics & analysis	asymptoticAnalysis, algoritmeAnalyseIntro, invariantSlides	O , Ω , Θ , o , ω comparisons; analysing loop run-times by hand.
Recurrences	divideAndConquer*, strassen	Master Theorem (CLRS 4th ed.) on $T(n) = aT(\frac{n}{b}) + f(n)$.
Sorting	sortingSlides, sortingInLinearTime	PARTITION /quick-sort, merge-sort, heap-sort; linear-time: counting sort, radix sort, bucket sort; stability & bounds.
Heaps / PQ	pqSlides	BUILD-MIN/MAX-HEAP , heapify, where keys can sit, extract/insert.

Hashing / dictionaries	dictionarySlides	Chain- ing; open ad- dress- ing; linear prob- ing & double hash- ing ; re- con- struct- ing in- sertion order.
Search trees	augmentedBSTSlides	BST opera- tions; red- black trees : validity (colour- ing), in- sertion + ro- tations; aug- mented trees.
Greedy	greedySlides	Huff- man coding (build tree, code- word lengths, total bits); ex- change- argu- ment reason- ing.
Dynamic programming	dynamicProgrammingSlides, maxSumSlides	Recur- rence $\rightarrow$ table;

		runtime & space of a DP; memoi- sation.
Graph traversal	graphTraversalSlides, connectedComponentsSlides	BFS (d- values, order), DFS (dis- covery/ finish times, edge classifi- cation: tree/ back/ for- ward/ cross), topo- logical sort .
Minimum spanning tree	mstSlides	Kruskal & Prim ; which edge enters next; #com- ponents during the run.
Shortest paths	shortestPathsSlides	Dijk- stra (no neg. edges), Bell- man- Ford (handles neg., de- tects neg. cy- cles), DAG- short- est-

		paths ; which algorithm is valid for a given graph.
Disjoint sets	<code>disjointSetsSlides</code>	Union-Find forest, union-by-rank + path compression ; resulting structure after a sequence.

3 Topic value ranking (what to study, in order)

Ranked by **return on investment** = points available $\times$ how mechanical/reliable the method is in a one-week window. Point estimates are from the June 2025 paper and are representative of the format.

3.1 Tier 1 — Do these first (high points, mostly mechanical)

1. **Asymptotic analysis, recurrences & loop analysis** — ≈ 26 pts.

The biggest single bucket: Master Theorem questions, $O/\Omega/\Theta/o/\omega$ true-false sets, and “what is the Θ -runtime of this loop”. Learn the Master Theorem cases and a handful of growth-rate facts once, and these are nearly automatic. **Best ROI in the entire course.**

2. **Discrete Mathematics block** — ≈ 25 pts.

Small, self-contained, and almost pure pattern-matching once definitions are memorised (logic equivalences, quantifier order, relation properties, induction shape). Don’t let the 20% weighting fool you — per hour studied this is as valuable as Tier 1. Make a one-page definitions sheet (allowed aid).

3. **Graph traversal: BFS, DFS, topological sort** — ≈ 17 pts.

Highly mechanical if you follow CLRS conventions (alphabetical adjacency lists, discovery/finish times). Edge classification and topological order are recurring earners.

3.2 Tier 2 — Strong value, learn next

1. **Sorting** (quicksort/PARTITION, counting sort, radix sort) — ≈ 13 pts.

Trace-the-array questions. Practise PARTITION and one pass of radix/counting sort by hand until reliable.

2. **MST + Disjoint sets** (Kruskal/Prim, Union-Find) — ≈ 9 pts.

“Which edge enters next”, “#components”, and union-by-rank/path-compression forests. Procedural and quick to drill.

3. **Greedy / Huffman** — ≈ 7 pts. Build the tree, count codeword bits, total encoded size. One reliable procedure.

4. **Red-black trees** — ≈ 7 pts. Validity-of-colouring and insertion. A bit fiddlier (rotations + recolouring) — worth it, but after the cheaper wins.

3.3 Tier 3 — Finish if time remains

1. **Heaps / priority queues** — ≈ 6 pts. BUILD-HEAP + heapify traces.

2. **Dynamic programming** — ≈ 6 pts. Reading runtime/space off a given recurrence.

3. **Hashing** — ≈ 5 pts. Linear probing & double hashing table reconstruction.

4. **Shortest paths** — ≈ 4 pts. Mostly “which algorithm is valid here?” (neg. edges $\Rightarrow$ not Dijkstra; cycles $\Rightarrow$ not DAG method). Cheap conceptual points.

If you only had two days: Tier 1 alone (asymptotics + discrete maths + graph traversal) is already **≈ 68 pts (>half the exam)** and is the most mechanical material — that’s your safety net.

4 Suggested one-week plan

Day 1	Asymptotics & recurrences (Tier 1.1). Memorise Master Theorem cases; drill $O/\Omega/\Theta$ comparisons and loop-runtime questions from past exams.
Day 2	Discrete Maths (Tier 1.2) — all three chapters. Build your allowed-aid one-pager: logic equivalences, quantifier negation, relation properties, induction template.
Day 3	Graph traversal (Tier 1.3): BFS, DFS times + edge types, topological sort. Trace several graphs by hand.

Day 4	Sorting + MST/Disjoint sets (Tier 2). Hand-trace PARTITION , radix/counting sort, Kruskal, Union-Find.
Day 5	Greedy/Huffman + Red-black trees (Tier 2). Then Heaps (Tier 3).
Day 6	DP, Hashing, Shortest paths (Tier 3). Light touch — aim for recognition, not mastery.
Day 7	Full timed mock exam using June 2025 (and 2023/2021), then review every miss. Finalise your cheat-sheet of procedures.

5 Practice material in this repo

- **Most representative:** `mcq-exams/SE4-DMAD-Eksamen-Juni-2025.pdf`, `...-2023.pdf`, `...-2021.pdf` — the current combined MCQ format. **Do these under timed conditions.**
- **Older DM507 MCQ exams** (2015–2019) — algorithms only, but same question style; good extra drill.
- **written-exams/ (dm507 2008–2016)** — older *written* format. Useful for understanding the algorithms deeply, but the current exam is MCQ, so treat these as study aids, not format practice.
- **Discrete maths:** `discrete-maths/old-exam-sets/` (DM547 + the SE4-DMAD June 2024 excerpt) and `exercises/opgaverDM01–09.pdf`.
- **Slides:** `algorithms-datastructures/slides/` (per the table above) and `discrete-maths/slides/Noter1–9.pdf` (Noter 1 is itself a nice topic map).

Point figures are estimates from the June 2025 paper and indicate relative weight, not a guarantee for your exam.